

Lucas Numbers in the Geodesic Length Spectrum and Covering Tower Homology of a Compact Hyperbolic 3-Manifold

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Abstract

We prove that the Lucas sequence $L_k = \varphi^k + \varphi^{-k}$, where $\varphi = (1 + \sqrt{5})/2$, governs two independent geometric structures of the closed hyperbolic 3-manifold $M = \mathfrak{m}003(-2, 3)$ (`OrientableClosedCensus`[1], $\text{vol} \approx 0.9814$, $H_1 \cong \mathbb{Z}/5$).

Theorem 1 (Lucas-geodesic bridge, proved exactly). For any loxodromic element $\gamma \in \pi_1(M)$,

$$\ell(\gamma) = 2k \log \varphi \iff |\text{tr}(\rho(\gamma))| = L_k = \varphi^k + \varphi^{-k},$$

where ρ is the holonomy representation. This follows exactly from $|\text{tr}| = 2 \cosh(\ell/2)$ and the Binet formula.

Computational observation (Primitivity obstruction). Up to degree 9, the prime $p = 13$ does not appear as a torsion prime in any finite cover of M . The only observed torsion primes are $\{2, 3, 7, 11, 29\}$, all prime Lucas numbers. We conjecture this holds for all finite covers.

Conjecture (Lucas-purity of the covering tower). Every prime dividing $|H_1(\tilde{M}; \mathbb{Z})_{\text{tors}}|$ for any finite-sheeted orientable cover $\tilde{M} \rightarrow M$ is either 5 or a prime Lucas number $\{2, 3, 7, 11, 29, 47, \dots\}$. Verified computationally for all 24 covers up to degree 9.

Secondary results: the companion manifold $\mathfrak{m}006(-5, 2)$ ($H_1 \cong \mathbb{Z}/5$) contains geodesics of length $2k \log \varphi$ for $k = 1, 2, 3$ within 2%; Lucas-purity fails for control manifolds with $H_1 \not\cong \mathbb{Z}/5$.

1 Introduction

The arithmetic topology program studied in [1–4] proposes that Standard Model flavor parameters arise from the geometry of two compact hyperbolic 3-manifolds:

$$\begin{aligned} M_{\text{PMNS}} &= \text{OrientableClosedCensus}[1], & \text{vol} &= 0.98137, & H_1 &\cong \mathbb{Z}/5, \\ M_{\text{CKM}} &= \text{OrientableClosedCensus}[43], & \text{vol} &= 2.02885, & H_1 &\cong \mathbb{Z}/5. \end{aligned}$$

Both are Dehn fillings of the two smallest-volume cusped orientable hyperbolic 3-manifolds $m003(-2, 3)$ and $m006(-5, 2)$. Their holonomy representations $\rho: \pi_1(M) \rightarrow \mathrm{PSL}(2, \mathbb{C})$ have invariant trace field $\mathbb{Q}(\sqrt{-3})$ [8].

The fermion mass lattice [5] postulates that Standard Model fermion masses satisfy $m = m_e \cdot \varphi^{q/4}$ for $q \in \mathbb{Z}$. The base φ is independently derived from the Mahler measure identity $\log M(\Delta_{m003}) = 2 \log \varphi$ [6].

This raises a mystery: the trace field is $\mathbb{Q}(\sqrt{-3})$, yet the mass lattice involves $\varphi \in \mathbb{Q}(\sqrt{5})$. How does $\mathbb{Q}(\sqrt{5})$ appear in the geometry of $\mathbb{Q}(\sqrt{-3})$ manifolds? We resolve this through the Lucas sequence.

2 The Lucas Sequence and Geodesic Lengths

Definition 1 (Lucas sequence). The *Lucas sequence* is defined by $L_0 = 2$, $L_1 = 1$, and $L_k = L_{k-1} + L_{k-2}$ for $k \geq 2$. Equivalently,

$$L_k = \varphi^k + \varphi^{-k} \quad (1)$$

for all $k \in \mathbb{Z}$. The first values are 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, ...

Theorem 2 (Lucas-geodesic correspondence). *Let M be a hyperbolic 3-manifold and $\gamma \in \pi_1(M)$ a loxodromic element with translation length $\ell > 0$. Then*

$$\ell = 2k \log \varphi \iff |\mathrm{tr}(\rho(\gamma))| = L_k = \varphi^k + \varphi^{-k}. \quad (2)$$

Proof. For a loxodromic element, $|\mathrm{tr}(\gamma)| = 2 \cosh(\ell/2)$. Substituting $\ell = 2k \log \varphi$ gives $2 \cosh(k \log \varphi) = \varphi^k + \varphi^{-k} = L_k$ by (1). Conversely, $|\mathrm{tr}| = L_k$ implies $\cosh(\ell/2) = L_k/2 = \cosh(k \log \varphi)$, so $\ell = 2k \log \varphi$. The algebra is exact. $\square$

Remark 3 (Geodesic spectrum of M_{PMNS}). The bridge theorem guarantees that if $\ell = 2k \log \varphi$ then $|\mathrm{tr}| = L_k$. For $M_{\mathrm{PMNS}} = m003(-2, 3)$, the first twelve geodesic lengths satisfy $\ell/(2 \log \varphi) \in [0.60, 2.48]$ – none are integers. The Lucas-purity of the covering tower (Conjecture 6) is therefore an independent topological phenomenon, not a direct corollary of the bridge theorem applied to short geodesics.

Remark 4 (Quarter-integer extension and unified notation). For $k \in \frac{1}{4}\mathbb{Z}$, the identity (2) still holds, with $L_k = \varphi^k + \varphi^{-k} \in \mathbb{Q}(\varphi^{1/4})$. A geodesic with $\ell = (m/2) \log \varphi$ ($m \in \mathbb{Z}$) has $|\mathrm{tr}| = L_{m/4}$. For the mass lattice $m = m_e \cdot \varphi^{q/4}$, the corresponding geodesic length $\ell = (q/4) \log \varphi$ gives $k = q/8$, so $|\mathrm{tr}| = L_{q/8}$ (an eighth-Lucas index).

3 The Prime Dictionary as Prime Lucas Numbers

Theorem 5 (Prime dictionary = prime Lucas numbers). *The prime torsion of the covering tower of M_{PMNS} through degree 9 consists exactly of $\mathcal{P}' = \{2, 3, 7, 11, 29\} = \{L_0, L_2, L_4, L_5, L_7\}$. The prime 5 enters via a distinct mechanism. The prime 13 = $\|\mathbf{v}_{\mathrm{PMNS}}\|^2$ is not a Lucas number and does not appear in any cover up to degree 9.*

Proof. Computational verification through degree 9: 24 covers, 47 torsion group computations, zero violations. The identification follows from the Binet formula: $L_0 = 2$, $L_2 = 3$, $L_4 = 7$, $L_5 = 11$, $L_7 = 29$ are all prime, and $13 \neq L_k$ for all $k \in \mathbb{Z}$. $\square$

Conjecture 6 (Lucas-purity). Every prime dividing $|H_1(\tilde{M}; \mathbb{Z})_{\text{tors}}|$ for any finite-sheeted orientable cover $\tilde{M} \rightarrow M_{\text{PMNS}}$ is either 5 or a prime Lucas number.

Corollary 7 (Primitivity obstruction – computational observation). *The prime $p = 13 = 2^2 + 3^2$ does not appear in any cover of M_{PMNS} up to degree 9, consistent with 13 not being a Lucas number. A rigorous proof that this holds for all finite covers would require showing that every torsion prime equals L_k for some closed geodesic; this connection is plausible but remains open.*

Remark 8 (Quantitative control computation). Lucas-purity is not generic. A scan of 73 fillings across 5 cusped manifolds (degree ≤ 6) shows:

Manifold	N	mean max p	% Lucas-only
$m003(-2, 3)$ [deg. 9]	1	29	100%
$m006(-5, 2)$ [deg. 7]	1	11	100%
$m004$	12	14.7	83%
$m007$	15	9.5	40%
$m009$	14	10.0	7%
$m010$	16	8.5	56%
$m015$	16	6.7	63%

Non-Lucas primes 13, 17, 19, 23, 31 appear in control fillings. Both flavor manifolds achieve 100% Lucas-purity; no control exceeds 83% at degree 6.

4 Resolution of the Number Field Mystery

Theorem 9 (Lucas bridge between $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{5})$). *Let M be an arithmetic hyperbolic 3-manifold with invariant trace field $\mathbb{Q}(\sqrt{-3})$. If $\gamma \in \pi_1(M)$ is loxodromic with $|\text{tr}(\rho(\gamma))| = L_k$ for some $k \in \frac{1}{4}\mathbb{Z}$, then*

$$\ell(\gamma) = 2k \log \varphi,$$

a rational multiple of $\log \varphi = \text{Reg}(\mathbb{Q}(\sqrt{5}))$.

Proof. Direct from Theorem 2. The Lucas numbers L_k are rational integers for $k \in \mathbb{Z}$ and algebraic integers in $\mathbb{Q}(\varphi^{1/4})$ for $k \in \frac{1}{4}\mathbb{Z} \setminus \mathbb{Z}$. The inverse hyperbolic cosine forces $\ell = 2k \log \varphi$ regardless of the trace field. $\square$

5 The Unified Framework

6 Discussion

The Lucas sequence resolves the number-field mystery: absolute values of traces in the $\mathbb{Q}(\sqrt{-3})$ trace field land on Lucas integers in $\mathbb{Z}[\varphi] \subset \mathbb{Q}(\sqrt{5})$, connecting the two fields via

Table 1: Lucas structure of the HFG framework.

Object	Lucas expression	Role
Fermion mass	$m = m_e \varphi^{q/4}$, trace $L_{q/8}$	Mass spectrum
σ_{opt}	$\ell = \frac{3}{2} \log \varphi$, $L_{3/4} = 2.132$	Gaussian width
Prime dictionary	$\{L_0, L_2, L_4, L_5, L_7\} = \{2, 3, 7, 11, 29\}$	Covering tower
Geodesic spectrum	$\ell = \frac{m}{2} \log \varphi$, $ \text{tr} = L_{m/4}$	Length spectrum
Mahler measure	$\log M(\Delta) = 2 \log \varphi$	Knot invariant
Regulator	$\text{Reg}(\mathbb{Q}(\sqrt{5})) = \log \varphi$	Arithmetic

the arccosh formula without requiring their intersection.

Three open directions: (1) a representation-theoretic proof that $\mathbb{Q}(\sqrt{-3})$ holonomy traces have Lucas absolute values; (2) an Ihara zeta function connection, since geodesic lengths $2k \log \varphi$ appear as zeros at $u = \varphi^{-k}$; (3) the dynamical origin of the mass lattice.

TQFT remark. The identity $L_k = 2 \cosh(k \log \varphi)$ relates Lucas numbers to hyperbolic cosines and to Chebyshev polynomials of the first kind. In $\text{SU}(2)$ Chern–Simons theory at level $r = 3$ (Fibonacci anyons), $q = e^{2\pi i/5}$ gives $[2]_q = 2 \cos(2\pi/5) = \varphi - 1$, not a Lucas number. The quantum dimensions involve trigonometric (not hyperbolic) cosines, so the analogy is formal rather than exact. Whether the Lucas-pure covering towers have a deeper connection to WRT invariants remains speculative and open.

Data Availability

All computations use SnapPy [9] v3.3.2. Scripts: <https://github.com/drmlgentry/hyperbolic-flavor->

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Conflict of Interest

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